Name: Anuttar Jain

Roll No: 22B0003

Indian Institute of Technology Bombay

PH 544 - General relativity **In Class Quiz 2** **Date 02-02-2026**

---

1. Observer O sees two events occur at the same place ($\Delta x = 0$) and separated in time by $\Delta t = 10^{-6}$ s. A second observer O' moving parallel to the x-axis sees them separated by $\Delta t' = 2 \times 10^{-6}$ s.

   (a) What is the speed of O' relative to O.

   (b) What is the separation in space of the two events according to O'?

2. Suppose a spaceship heading directly towards the Earth at half the speed of light sends a signal that leaves the ship at speed c as observed from the ship. Calculate the speed at which the signal approaches the Earth.

Q1.

$(x, t) \to O$ frame

$(x', t') \to O'$ frame.

(a) $\Delta t' = \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right)$

$\Delta t = 10^{-6}$, $\Delta x = 0$, $\Delta t' = 2\times10^{-6}$

$\Rightarrow \quad \gamma = 2 \quad \Rightarrow \quad \boxed{v = \frac{\sqrt{3}}{2}c}$ ✓ 1

✓ 1

Soln

(b) $\Delta x' = \gamma(\Delta x - v\Delta t) = -v\gamma\Delta t$

$\Rightarrow \Delta x' = -\sqrt{3}\times10^{-6}\,c = -300\sqrt{3}$ m ✓ 1 $\frac{3}{3}$

Q2. Light always travel at speed c as stated by Ist Law of STR.

$\therefore$ speed of approach of signal to Earth $= c$ ✓

Mathematically,

Let (x, t) be in spaceship frame, Let (x', t') be in Earth frame.

$$\therefore \frac{dx'}{dt'} = \frac{\gamma\left(\frac{dx}{dt} - v\right)}{\gamma\left(1 - \frac{v}{c^2}\frac{dx}{dt}\right)} = \frac{c\left(1 - \frac{v}{c}\right)}{\left(1 - \frac{v}{c}\right)} = c$$ ✓

✓ ✓ $\frac{2}{2}$ $\frac{5}{5}$